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Conventional or Rotary Steerable for Shale Well Construction? - A New Steerable Motor with Unique Features Designed to Fill the Economic Gap Between Conventional and Rotary Steerable

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Abstract

Shale drilling applications often expose the economics gap between conventional and rotary steerable systems (RSS). Well budget constraints determine the selection process between the two options. An unexpected trip due to RSS reliability or directional performance can mean the difference between meeting the budget or exceeding it. A new steerable motor has been designed with features specifically for shale drilling applications, providing an economic alternative to RSS.

The new steerable motor is constructed with a unique bit-deflection mechanism that allows for higher surface rotation speeds (like RSS) while rotary drilling. While slide drilling (steering), the new steerable motor can generate higher dogleg severity than a conventional bent-housing steerable motor by utilizing a lower-angle bent housing. The new steerable motor also contains an instrumentation package that is different from anything in the past. The instrumentation package is self-contained and mounted in the centerline of the motor. Real-time drill-bit dynamics, near-bit inclination and pseudo gamma measurements are available as an option on the new steerable motor, providing geo-steering and drill-bit performance information to enhance shale drilling operations.

This paper will explain the directional performance and real-time measurements from wells drilled in North America land utilizing the new steerable motor. We will show examples of where the new steerable motor provides economic benefits compared to conventional steerable motors and RSS. Conclusions will be drawn on the technical and economic benefits of the new steerable motor, specifically addressing the gap that exists between conventional steerable motors and RSS.

The new steerable motor is innovative because of its bit deflection method. This simple design provides drilling benefits compared to conventional steerable motors. The onboard real-time drilling dynamics and geo-steering packages are unique.

North America Shale Drilling Landscape "a la carte"

The North America shale drilling landscape has changed significantly. From a directional drilling service provider's perspective, gone are the days when one service provider supplied all the products and services, such as RSS, motors, measurement-while-drilling (MWD) and directional services. Today, the offering is "a la carte", and operators are utilizing RSS, motors, MWD and directional services from multiple suppliers to deliver the optimal solution for performance, reliability and cost. The "a la carte" option allows the operators to select the best performing products and services for their applications.

The RSS market has also changed with the major service companies now offering their RSS for direct sale in North America. While these major service companies still maintain control of the repair and maintenance (R&M) of the RSS, directional companies can now own these RSS as their own assets. With the two largest service companies offering their RSS for direct sale in North America, competition is becoming fierce. There are still small independent private technology companies developing RSS for the North America shale market, and these will bring even more competition to the North America shale drilling industry.

There are a significant number of mud motor and power section suppliers in North America, making the competition strong.

Independent MWD suppliers are also developing some of the best and most reliable MWD technology for shale drilling. One of these MWD suppliers has captured a significant percentage of the rigs, with strong reliability, service and high-temperature offerings.

Directional providers supplying directional drillers with third-party rental tools need little more than a pick-up truck and a cell phone to get started in business. They rent all their tools from a multitude of suppliers in North America. One limitation is that they lose control of product service quality as they are dependent on third-party suppliers. Some operators have brought their own directional services in-house, along with remote operating pods that oversee the rigs.

North America Shale Well Construction

Shale well construction in North America consists of lateral and U-lateral designs. Both these well construction designs have curves to land planned at 8 to 12 degrees per hundred feet (Schultz and Kiefner 2022).

Lateral design lengths are getting longer and typically range from two-miles (10,560ft.) to three-miles (15,840ft.). Today, operators are inching towards four-mile (21,120ft.) laterals. Fig. 1 shows the average lateral length by basin over the years (Mayer 2024).

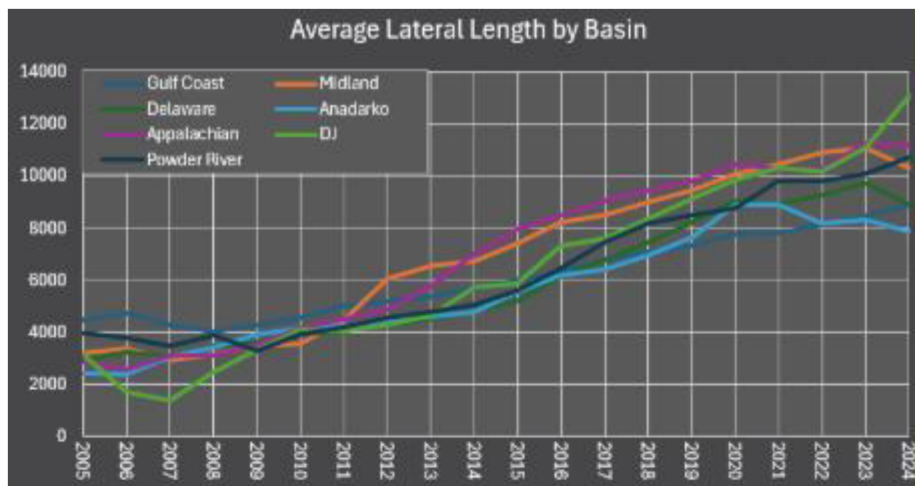


Figure 1—Average Lateral Length by Basin (www.tgs.com/weekly-spotlight/01-06-2025).

U-lateral designs are becoming more common, specifically in locations where engineers are confronted with "stranded acreage". The first known horseshoe lateral was the Neelie #4H well drilled by Shell Oil in the Delaware Basin in 2019 (Feller 2024). Since then, U-lateral designs have become more common because of increased rate of return (ROR). In the Delaware Basin, where the most U- laterals have been drilled, a 25-40% increase in ROR is realized by replacing one-mile laterals with U-laterals. Moving forward, one operator has plans for 14 U-laterals in one section in the Midland Basin. The 8-mile development takes the industry from U-lateral curiosity to a serious stranded section development strategy (U Driller 2024). Fig. 2 shows U-lateral design and terminology.

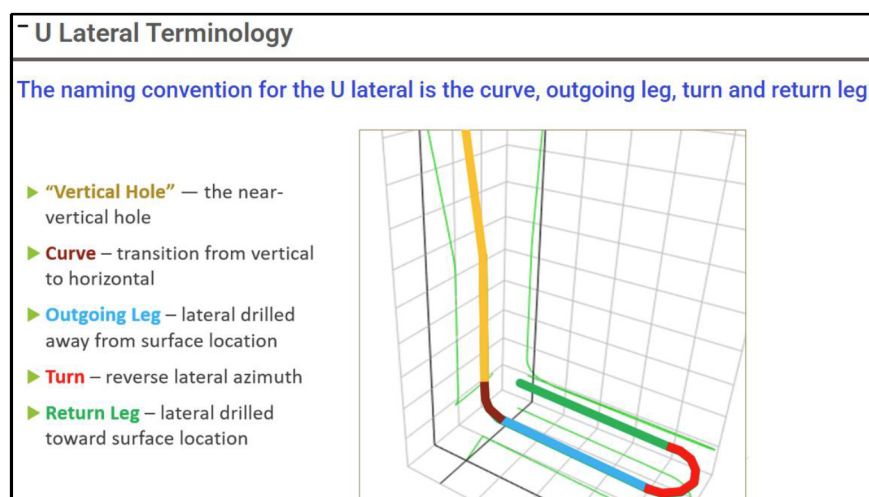


Figure 2—U-lateral design and terminology (udriller.com/economics/).

Choice between Conventional Steerable Motor or RSS

The choice between a conventional steerable motor or a motor-driven RSS depends on the wellbore construction design, basin, bench, technical requirements and costs. RSS delivers technical advantages such as a faster rate of penetration (ROP) due to 100% rotation, improved wellbore quality, improved torque and drag, less skilled directional drillers required (easier to enable remote operations), and near-bit real-time measurements (inclination, gamma and shocks and vibrations). However, all of this comes with a higher operating cost and higher risk of downhole failure on sensitive RSS components.

In some cases, to optimize economics, steerable motors may be used to drill the first portion of the well, with motor-driven RSS being picked up later in the well because of torque and drag requirements (especially for three- and four-mile laterals).

The main operational limitations of a steerable motor are:

- A large bend angle is required to build a high-dogleg curve, which limits maximum surface RPM when rotating in the lateral.
- Sliding ROP is typically slower than rotating ROP; therefore, a neutral rotary response is preferable.
- No built-in real-time near-bit measurements, such as inclination, gamma and drill-bit dynamics.

There was clearly a need to find a solution for the limitations of the conventional steerable motor and develop a product that would fit this middle ground below RSS at a competitive cost.

New Steerable Motor Mechanical Construction

A new steerable mud motor has been developed specifically for shale drilling applications (Jones and Sugiura 2021). The requirements of the new steerable motor were to develop a product that would fill the economic and technical gap between RSS and steerable motor technology.

The new mud motor has been constructed to provide optimized geometry to improve dogleg capability with lower bend angle, higher surface RPM capability while rotary drilling and ideal stabilizer touch points for a neutral rotation tendency in the lateral (Jones et al. 2025).

Fig. 3 shows the new-generation steerable motor with internal tilted mandrel, external bent housing and optimized placement of stabilization and wear pad. This design is unique and has been specifically developed for shale drilling well construction to achieve vertical, curve and lateral in one run.

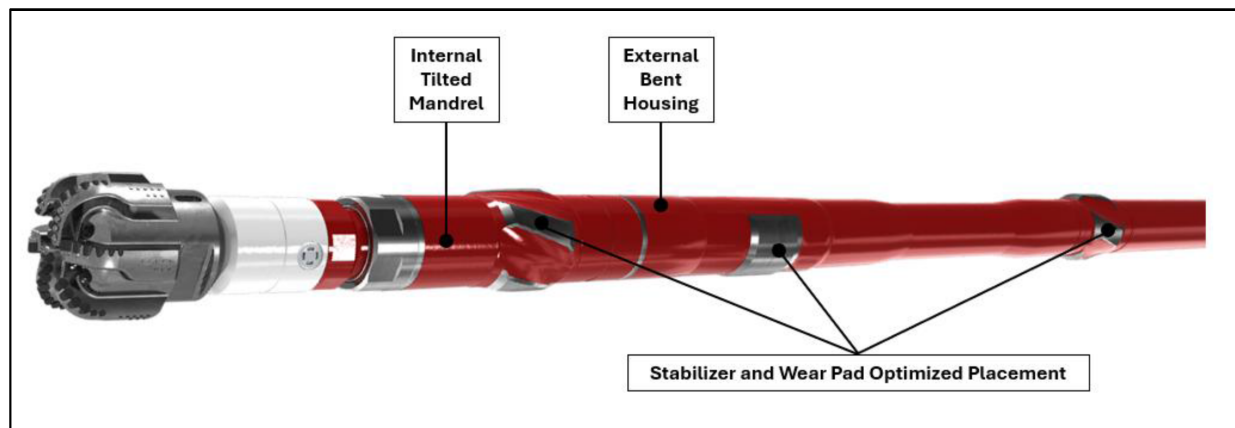


Figure 3—New Generation Steerable Motor Lower End (Jones et al. 2025).

New Steerable Motor Real-Time Measurements Hardware

To bridge the real-time measurements gap with RSS, the new steerable motor has onboard real-time measurements for drilling dynamics and geo-steering. The real-time measurements can be divided into two distinct fields:

1. Drill bit drilling dynamics
 - Drill bit lateral and axial shocks/vibrations
 - Drill bit rotation speed in revolutions per minute (RPM)
 - Drill bit stick-slip
 - Drill string rotation speed in RPM
 - Drill string stick-slip
 - High-Frequency Torsional Oscillations (HFTO)
 - Mud motor back-drive
2. Geo-steering
 - Near bit inclination
 - At-bit pseudo gamma

Fig. 4 shows the new-generation instrumented steerable motor bottom-hole assembly (BHA) with real-time short-hop to MWD. The universal bottom hole orientation (UBHO) sub repeater provides reliable real-time communication across MWD bottom-mount pulser.

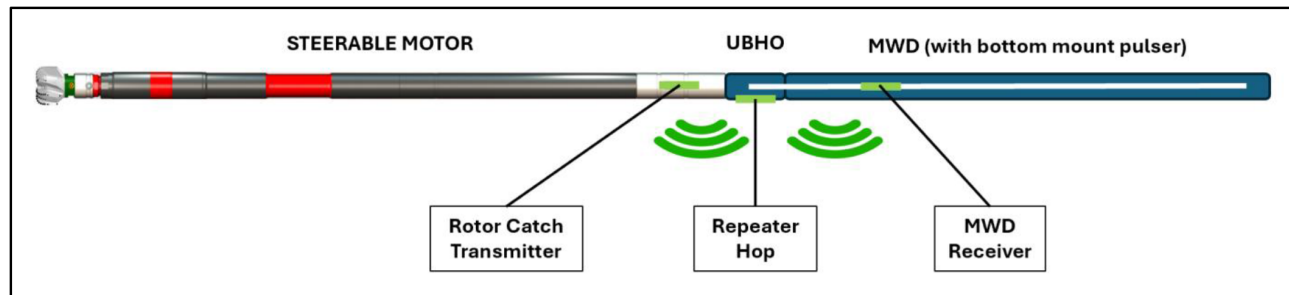


Figure 4—Instrumented Steerable Motor BHA with Real-time Short Hop to MWD (Excerpted from SPE-223765-MS).

To enable real-time measurements, the following components are required -

1. Instrumented Rotor Catch - Houses drilling dynamics, inclination, pseudo gamma and real-time transmitter.
2. UBHO Sub Repeater Antenna – Real-time transmission for MWD with bottom mount pulser.
3. MWD Receiver Antenna – Module with receiver antenna and drilling dynamics.

Fig. 5 shows the instrumented rotor catch assembly with real-time transmitter. The design is unique in that it does not interfere with the mechanical integrity of the drilling motor (Sugiura and Jones 2023).

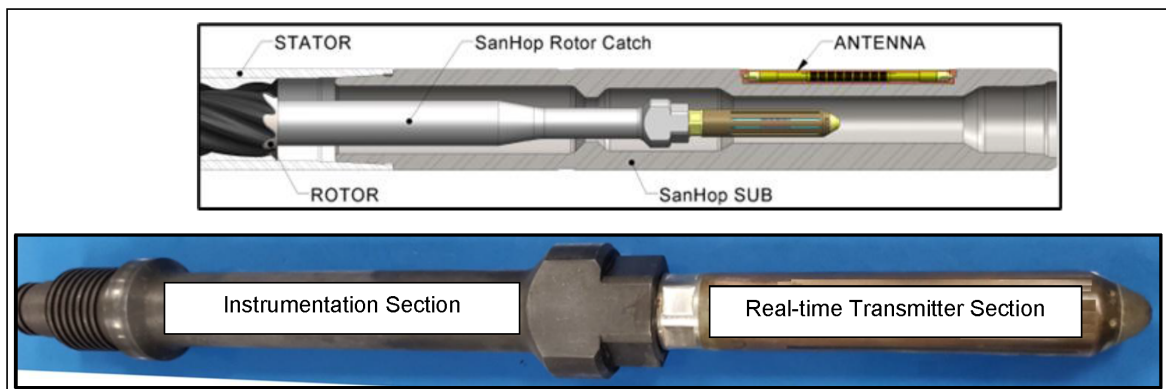


Figure 5—Instrumented Rotor Catch Assembly (Excerpted from SPE-213650-MS).

The instrumentation section contains solid-state accelerometers (shock and vibration), gyro (rotation speed), inclinometer, lithium battery and associated electronics (Jones and Sugiura 2019; Jones and Sugiura 2020).

Fig. 6 shows the UBHO sub repeater antenna. This allows real-time short hop communication across an MWD bottom-mount pulser. The system can also be utilized with MWD top-mount pulser, and this does not require the UBHO repeater antenna.

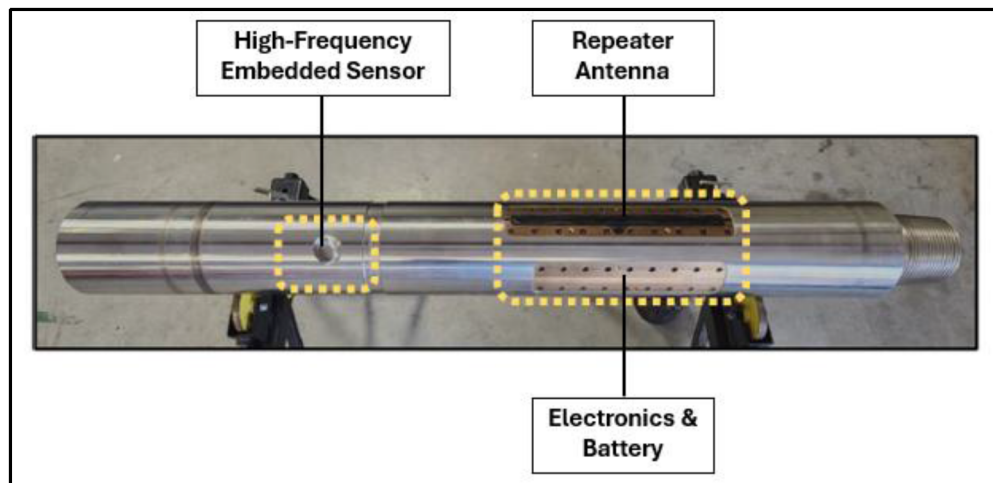


Figure 6—UBHO Repeater Antenna (Excerpted from SPE-223765-MS).

Fig. 7 shows the MWD receiver antenna module. The MWD receiver module also contains high-frequency drilling dynamics measurements specifically to identify mud motor back-drive dynamics.

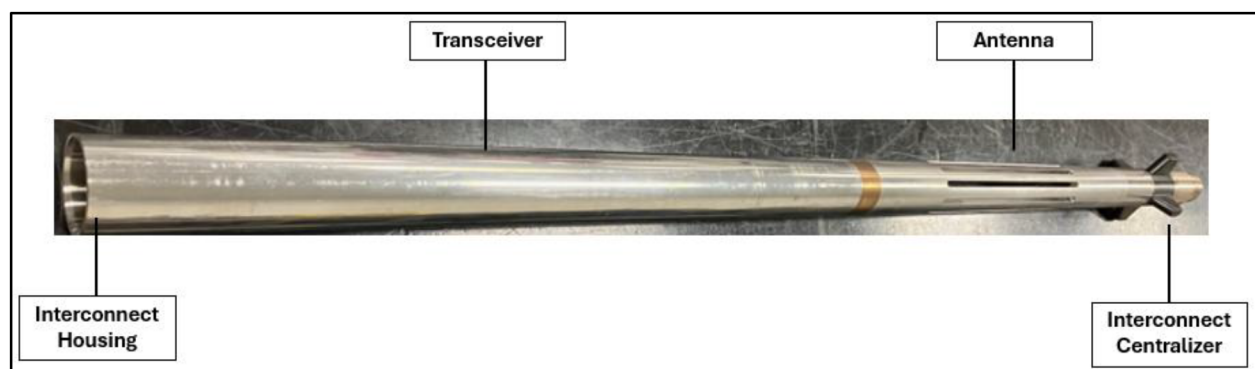


Figure 7—MWD Receiver Antenna Module (Excerpted from SPE-223765-MS).

New Steerable Motor Run Statistics

The new steerable motor has been commercially available in North America since January 2022. To date the motor has drilled over 17 million feet. Most of the footage has been drilled in the Permian Basin (Delaware and Midland Basins). 7.875" and 8.50" are the most common hole sizes; however, the motor is available in all hole sizes ranging from 6.125" to 17.50".

Fig. 8 shows the footage totals from 2022 to 2024.

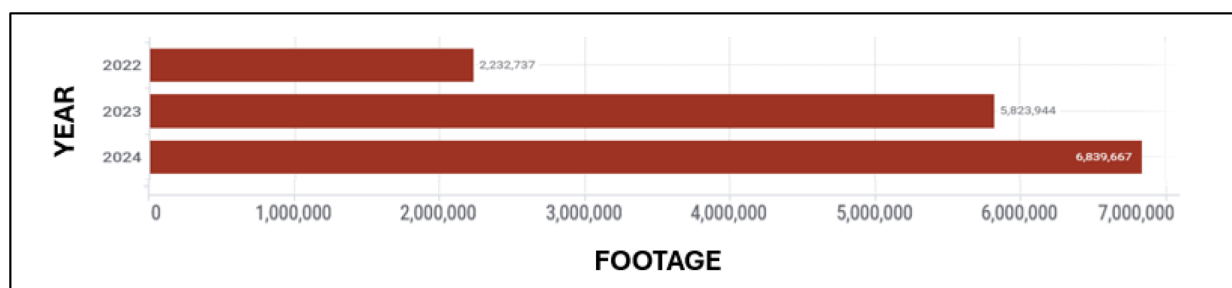


Figure 8—New Steerable Motor Footage Over the Years.

Operators are realizing the benefits of the new motor for drilling vertical-curve, curve-lateral or vertical-curve-lateral applications. The choice of application depends on well design and basin characteristics. Technical planning to select the best applications is determined by the criteria of eliminating a bit or BHA trip to improve cycle time economics.

Drilling performance gains come from higher surface rotation, less time sliding (less footage) and optimal stabilizer placement for neutral build/drop tendencies in the lateral.

New Steerable Motor Directional Performance Versus Conventional Steerable Motor

The build rate of the new steerable motor is higher than that of a conventional steerable motor in slick and stabilized configurations, due to its unique geometry. For example, in 8 1/2" hole size, a 1.75 bend angle slick new steerable motor will yield the same as a 1.95 bend angle slick conventional steerable motor. And a 1.50 bend angle stabilized new steerable motor will yield the same as a 2.25 bend angle stabilized conventional steerable motor.

The advantages of a lower bend angle include reduced bit orbit diameter, improved borehole quality, improved torque and drag, and the ability for higher surface RPM (similar to RSS), resulting in improved ROP while rotary drilling.

Fig. 9 shows the external bend angle and build rate comparison for conventional steerable motor and new steerable motor in 8 1/2" hole size.

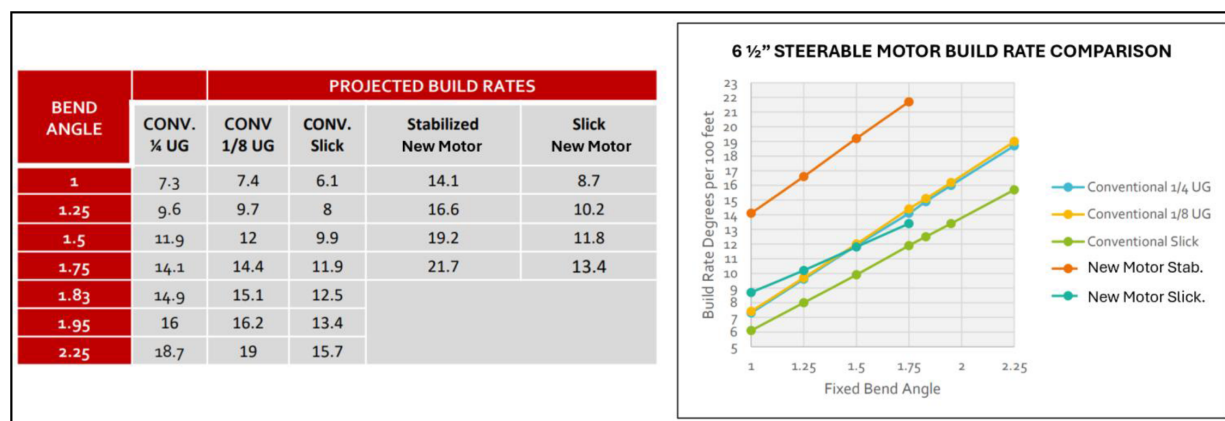


Figure 9—External Bend Angle and Build Rate for Conventional Steerable Motor and New Steerable Motor in 8 1/2" Hole Size.

A typical shale well construction design consists of three distinct sections, namely vertical, curve and lateral (referred to as VCL). To achieve all three sections, typically three different conventional steerable motor configurations are required. This requires two trips to get the correct configuration of motor to drill the well.

With the new steerable motor, one configuration can drill the vertical, curve and lateral portions of the well with no trips for different configurations. This aligns the new steerable motor with a one-run-to-TD approach, saving significant time and aligning the new motor with the capability of a high-dogleg RSS.

Fig. 10 shows the comparison of motor configurations required to drill vertical, curve and lateral.

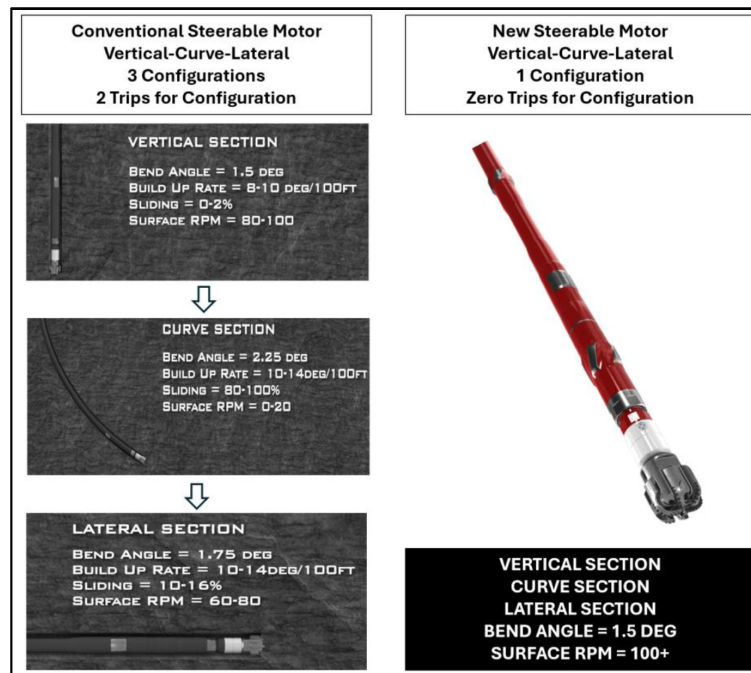


Figure 10—Comparison of Motor Configurations Required to Drill Vertical, Curve and Lateral.

As a comparison for the lateral section only, a conventional steerable motor typically utilizes a 1.75-degree bend angle, which limits the surface rotary speed to 60-80 RPM. With the new steerable motor, a 1.50-degree bend angle allows the rotary surface speed to be increased up to 120 RPM which improves hole cleaning and ROP, ultimately bringing the performance closer to an RSS.

Fig. 11 shows an example of days versus depth for a conventional steerable motor versus the new steerable motor.

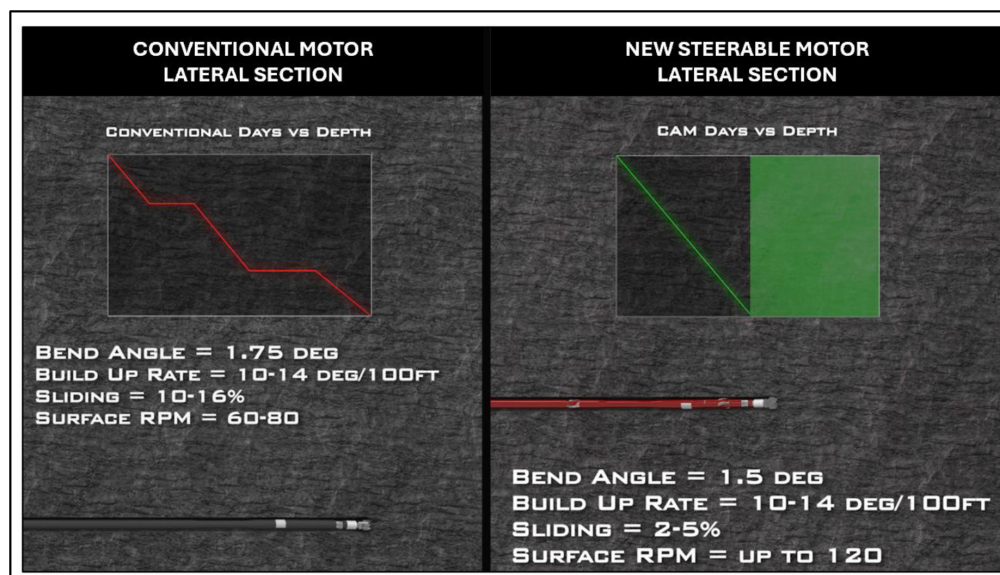


Figure 11—Example of Days Versus Depth for Conventional Steerable Motor Versus New Steerable Motor.

The unique geometry of the new steerable motor also enhances rotary performance. In benches where directional tendencies are strong, the stabilized version of the new steerable motor can be configured with a

bearing housing stabilizer and a stabilizer below the stator. This configuration brings all three touch points close together to neutralize the rotary tendency of the BHA.

Fig. 12 shows stabilized and slick versions of the new steerable motor.



Figure 12—Stabilized and Slick Versions of New Steerable Motor.

Orbit Diameter and Borehole Quality

RSS delivers improved borehole quality because there is no external bend. The assembly runs true with the wellbore and side force is applied from an actuator behind the bit. A steerable motor utilizes a fixed external bend; therefore, the larger the external bend, the greater the effect on orbit diameter and borehole quality.

The new steerable motor has the advantage of a lower angle external bend, which reduces the bit orbit diameter and provides improved borehole quality and bit life, bringing it closer to RSS performance.

Fig. 13 shows the difference in bit orbit diameter for a conventional steerable motor and the new steerable motor.

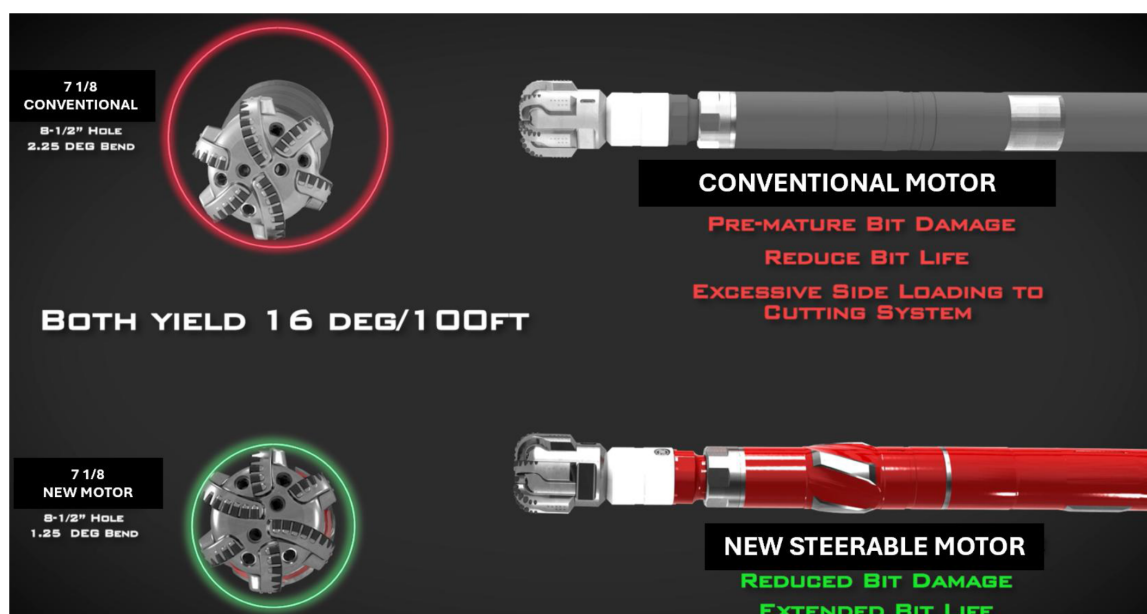


Figure 13—Bit Orbit Diameter Conventional Steerable Motor and New Steerable Motor.

New Steerable Motor Real-Time Measurement Examples

To align the new steerable motor with RSS capabilities, real-time measurements of inclination, pseudo gamma and drill-bit dynamics are available. This provides a significant differentiation over standard conventional steerable motors.

Fig. 14 shows the new steerable motor's real-time near-bit inclination while sliding to build the curve, compared to MWD dynamic inclination.

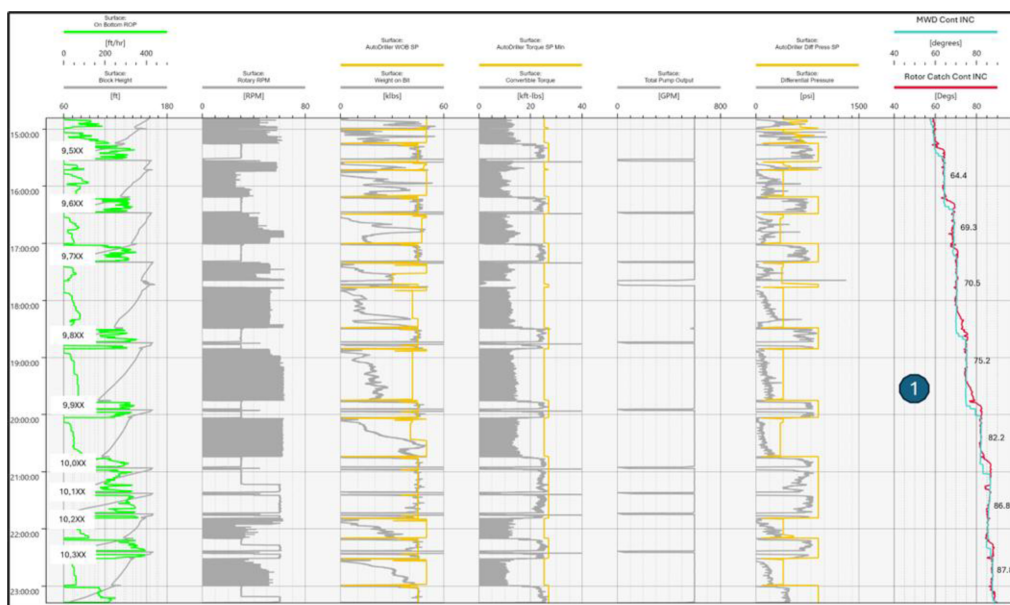


Figure 14—Inclination Measurements While Building Curve (Excerpted from SPE-223765-MS).

Fig. 15 shows the new steerable motor's real-time near-bit inclination while rotary drilling in the lateral compared to MWD dynamic inclination.

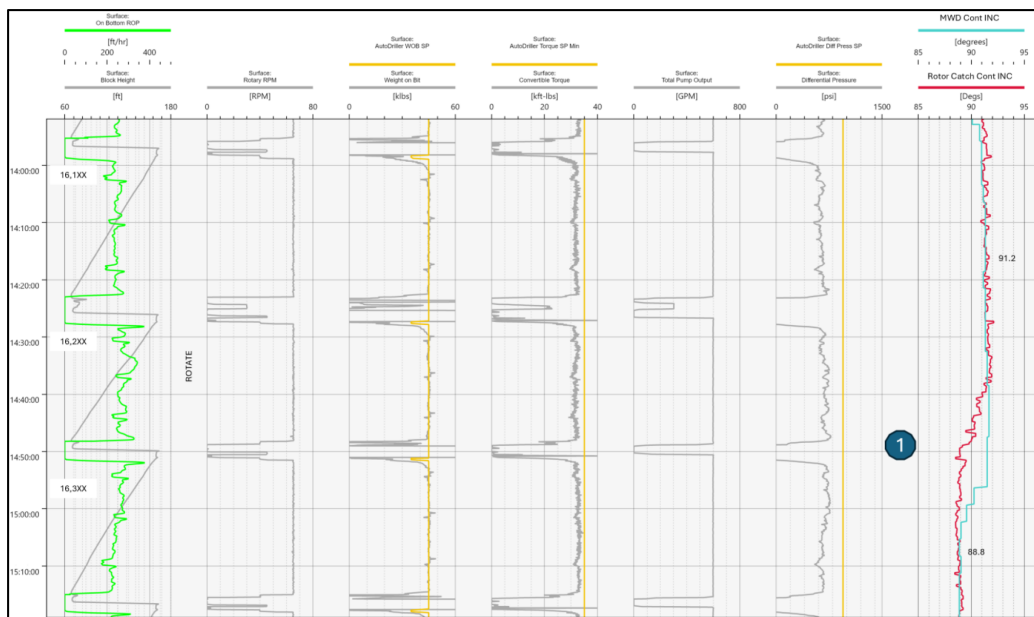


Figure 15—Inclination Measurements While Rotary Drilling Lateral (Excerpted from SPE-223765-MS).

Fig. 16 shows the new steerable motor's real-time pseudo gamma compared to MWD API gamma.

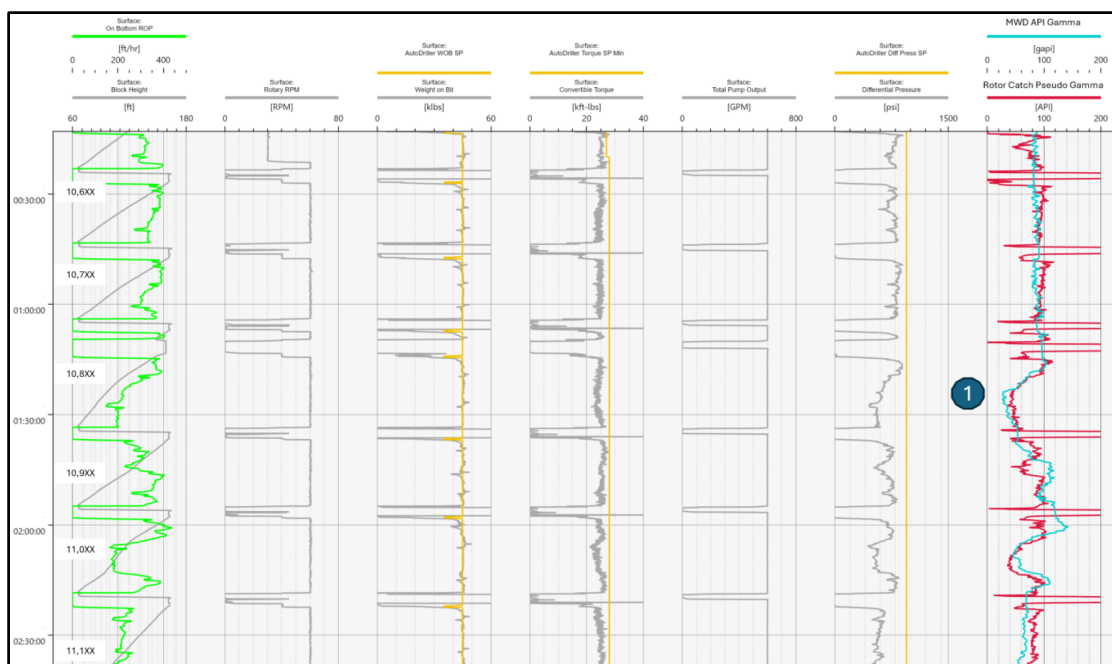


Figure 16—Rotor Catch Pseudo Gamma Versus MWD Gamma (Excerpted from SPE-223765-MS).

Fig. 17 shows the new steerable motor's real-time drill-bit dynamics compared to MWD shock and vibration (S&V).

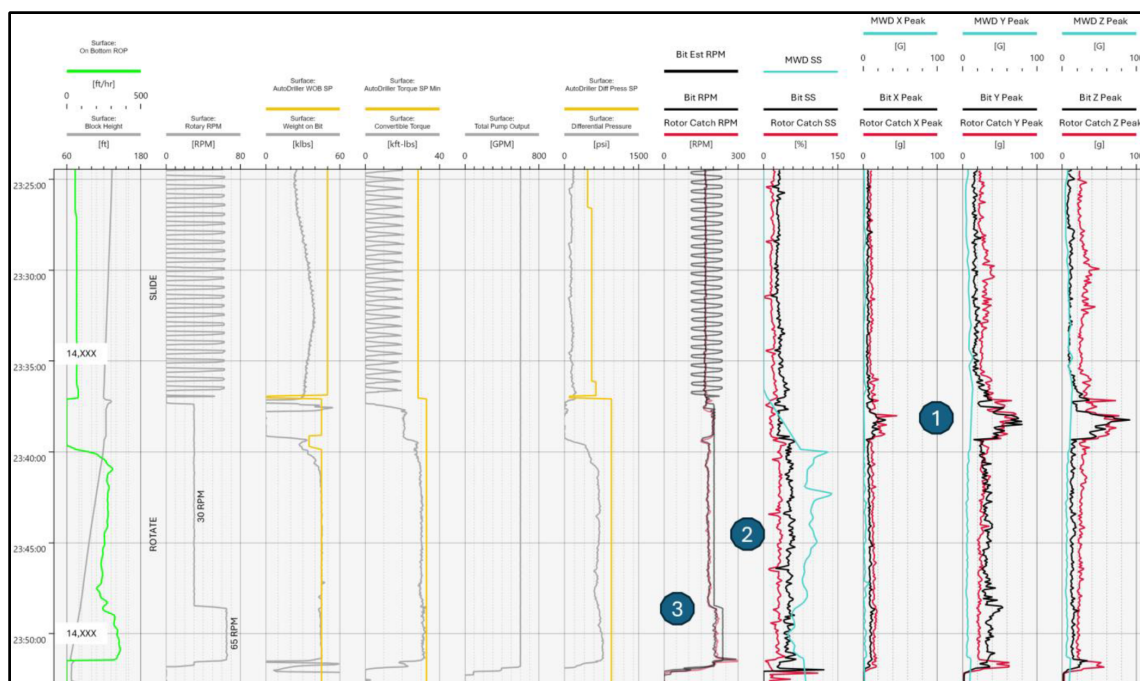


Figure 17—Slide-to-Rotate Transition (Excerpted from SPE-223765-MS).

Item 1 shows the increase in axial, tangential and radial shocks seen at the drill bit (black traces), corresponding with the rotor catch shocks (red traces). The MWD shocks (blue traces) do not show an increase. This is because the MWD shock measurements are placed above the mud motor, and not reflective of the shocks that are present at the drill bit. Utilizing the real-time drilling dynamics data from the new-

generation steerable motor allows the operator to optimize rig input RPM, flowrate and pick-up/back-to-bottom technique to minimize bit shocks and thereby improve drill bit/mud motor life.

Item 2 shows MWD stick-slip (blue trace) at a higher magnitude than bit (black trace) and rotor catch (red trace) stick-slip. This is because the drill string is turning at a lower RPM than the drill bit.

Item 3 shows that the bit RPM (black trace) and rotor catch RPM (red trace) are reading the same values. The RPM is lower than the theoretical estimated bit RPM because the motor is on-bottom and under load. The real-time RPM measurement from the new-generation steerable motor provides the operator with a bit RPM with no load (off-bottom) and under load (on-bottom) providing an indication of the work being done by the motor (and the strength of the motor over time – degradation).

The Economic Advantage Versus RSS

RSS economics can become marginal if there are any issues/limitations with directional performance, ROP, reliability and damages. In shale drilling, one trip at depth due to an RSS issue can have a negative effect on operating margins. A lost-in-hole (LIH) event with RSS/MWD/Motor can be even more serious, sometimes turning operating margins upside down.

Wells designs that have tight operating margins mean the decision between RSS or conventional steerable motors is crucial. The new steerable motor now fills this gap, by providing a lower-cost alternative to RSS, with improved directional performance and optional near-bit measurements.

When comparing the average cost of an RSS versus the new instrumented steerable motor, the daily operating rate of the new steerable motor results in a 50% saving on operating and service costs. Therefore, selecting the new steerable motor for the right applications can bring significant cost savings.

Obviously, RSS is necessary for 4-mile laterals, but if bit trips get in the way of performance, the new steerable motor for the curve and first portion of the lateral, and then switching to RSS for the second portion of the lateral to TD, may be the best economic choice.

Conclusions

The new steerable motor has proven to deliver a performance advantage over conventional steerable motors due to its higher sliding DLS yield with a lower external bend. The lower external bend angle allows for higher surface RPM while rotary drilling, improving ROP and hole cleaning capability. The stabilization options to optimize rotary build/drop tendency enable a balanced lateral BHA to maximize rotary-drilling time for highest ROP.

Onboard real-time measurements provide data comparable to what is seen from an RSS, allowing for geo-steering and optimization of drill-bit dynamics.

Bridging the technical and economic gap between RSS and steerable motors offers a new downhole tool for the drilling engineer.

Acknowledgements

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